

Foundation of Blockchain Technology (IS501)

Pre-Reading Material – Lecture 4

Proof of Work

Estimated Reading Time: 20–25 minutes

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

Blockchain is a decentralized system. Unlike a traditional database, there is no single authority that decides which transactions should be added to the ledger.

This creates an important question:

How can participants in a decentralized network agree on the next valid block?

Bitcoin addresses this problem using **Proof of Work (PoW)**. In PoW, miners perform computational work to earn the opportunity to add a new block to the blockchain. The computational requirement also makes it expensive to modify previously accepted blockchain data.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the basic idea of Proof of Work.
- Explain why Bitcoin requires a consensus mechanism.
- Describe the role of miners.
- Explain the relationship between **nonce, hash, and difficulty**.
- Explain how PoW contributes to blockchain security.
- Describe the basic idea of a **51% attack**.



Activate Your Prior Knowledge

Think About It

Imagine that thousands of computers maintain copies of the same Bitcoin ledger.

Question

1:

Who should get the right to add the next block?

Question

2:

How can the network select a participant without depending on a central authority?

Question

3:

If someone changes an old transaction, what happens to the block's hash?

Prerequisite Concepts

Before reading this topic, you should understand:

- Blockchain
- Blocks and transactions
- Distributed ledger
- Cryptographic hash functions
- Bitcoin

Core Concepts

1. Consensus and Proof of Work

A blockchain needs a **consensus mechanism** to allow participants to agree on the valid state of the blockchain.

Bitcoin uses **Proof of Work**.

In simple terms:

Proof of Work requires participants to perform computational work before they can add a block to the blockchain.

The participants performing this work are called **miners**.



The reference material describes PoW as a mechanism in which miners compete by solving a computational problem, with the successful miner gaining the privilege of writing to the ledger.

2. How Does Mining Work?

A simplified mining process is:

Transactions



Construct Block



Change Nonce



Calculate Hash



Does Hash Satisfy Difficulty?

No → Try Again

Yes → Broadcast Block

The miner repeatedly changes the **nonce** and calculates the block hash until a valid result is obtained.

Other nodes can then verify the proposed block.

3. Nonce, Hash and Difficulty

Nonce

A **nonce** is a value that miners change repeatedly while searching for a valid block hash.

For example:

Nonce	100	→	Invalid
Nonce	101	→	Invalid
Nonce	102	→	Invalid
...			
Nonce	874563	→	Valid

The miner generally has to perform many trials **because** the correct value cannot simply be predicted.

Hash

The block header is hashed using a cryptographic hash function.

A small change in the input produces a completely different hash.

Therefore, modifying transaction data changes the resulting hash.

Difficulty

The network specifies how difficult it should be to find a valid hash.

Higher difficulty → More computational work required

4. Why Is It Called "Proof of Work"?

The successful miner provides evidence that computational work has been performed.

The important property is:

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Course Repository: <https://github.com/rajesh101191/Foundation-of-Blockchain-Technology>



Finding a valid solution is computationally difficult, but verifying the solution is relatively easy.

This asymmetry makes PoW useful for a decentralized consensus system.

5. Proof of Work and Blockchain Security

Suppose an attacker changes a transaction in an old block.

The modification changes the block's hash.

The attacker would then need to redo the required computational work for that block and, depending on the attack, compete with the network's continuing chain.

Therefore, the computational cost makes arbitrary modification of the blockchain difficult.

The reference material specifically discusses the computational cost involved in altering the ledger and the role of PoW in protecting the blockchain.

6. The 51% Attack

A **51% attack** refers to a situation where an entity or group controls more than half of the network's mining/computational power.

With majority computational power, an attacker may be able to:

- Outpace honest miners.
- Reorganize portions of the recent blockchain.
- Attempt double-spending.

However, a 51% attack does **not** mean that the attacker can simply violate every rule of the Bitcoin protocol.

The key idea is:

PoW security depends strongly on the distribution of computational power across the network.

Proof of Work at a Glance

Concept	Meaning
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Consensus	Agreement about the valid blockchain state
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Proof of Work Consensus based on computational work

Miner	Participant performing mining computations
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Nonce	Value changed during the mining search
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Hash	Output of the cryptographic hash function
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Difficulty	Determines how difficult it is to find a valid solution
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Mining	Process of performing PoW to find a valid block
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51% Attack	Attack involving majority computational power
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Guided Reading Questions

1. Why does Bitcoin need a consensus mechanism?
2. What is Proof of Work?
3. What is the role of a miner?
4. Why does a miner repeatedly change the nonce?
5. What is the purpose of mining difficulty?
6. Why does changing a transaction affect the block?
7. What is a 51% attack?



Reflection Activity

Consider this situation:

A company controls 60% of the total mining power of a blockchain network.

In **100–150 words**, explain:

- Why this could be a security concern.
- What the company might potentially do.
- Why decentralization of mining power is important.

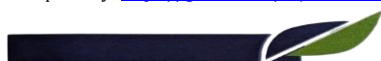
Self-Assessment

Fill in the blanks

1. Bitcoin uses _____ as its consensus mechanism.
2. Participants who perform mining computations are called _____.
3. The value repeatedly changed during mining is called the _____.
4. The output of a cryptographic hash function is called a _____.
5. The difficulty determines how _____ it is to find a valid solution.
6. An attack involving majority computational power is called a _____ attack.

True or False

7. Proof of Work requires computational resources.
8. Changing the transaction data does not affect the block hash.
9. Miners compete to find a valid solution to the PoW puzzle.
10. A 51% attack involves control of a majority of computational power.



Think Before Class

Question

Proof of Work provides security through computational work, but mining consumes computational resources and energy.

Do you think the security provided by PoW justifies its computational cost? Why or why not?

Be prepared to discuss your answer in class.

Key Takeaways

- Bitcoin uses **Proof of Work** as its consensus mechanism.
- **Miners** perform computational work to compete for the right to add a block.
- The mining process involves repeatedly changing a **nonce** and calculating hashes.
- **Difficulty** controls how difficult it is to find a valid solution.
- PoW makes modification of historical blockchain data computationally expensive.
- A **51% attack** occurs when an entity or group controls a majority of the network's computational power.

Reference Reading

Understanding Proof-of-Work — Fidelity Digital Assets, *Understanding Proof-of-Work*.

Additional

Reference:

Block Chain — Bitcoin Developer Documentation, particularly the **Block Headers** section for understanding Bitcoin's block-header hashing and Proof-of-Work process.

Recommended reading: Focus on **Proof of Work, Mining, Security of Proof of Work, and 51% Attack**

